Spatial data models

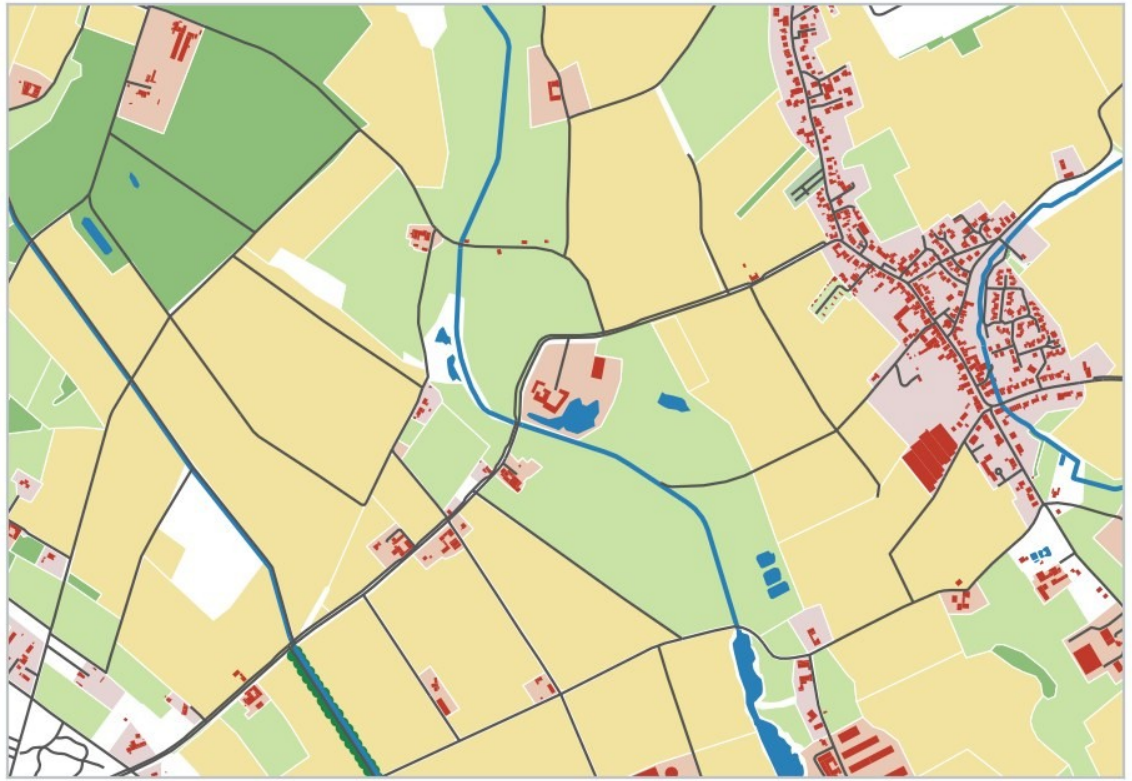
Vector, raster and the OGC Simple Features model

Examples: Kamp-Lintfort and Frohnenbruch



Same place, two models

Vector: discrete objects



Raster: continuous field of 10 m cells

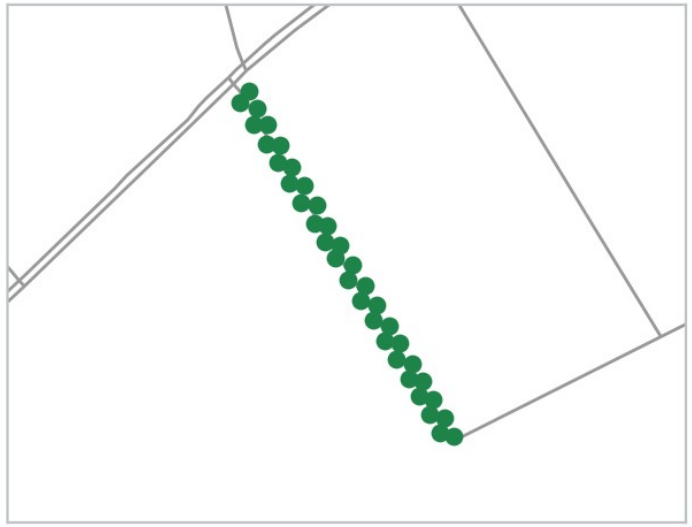


Vector models things that have a boundary and an identity. **Raster** models a quantity that varies everywhere in space.

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Vector model: geometry + attributes

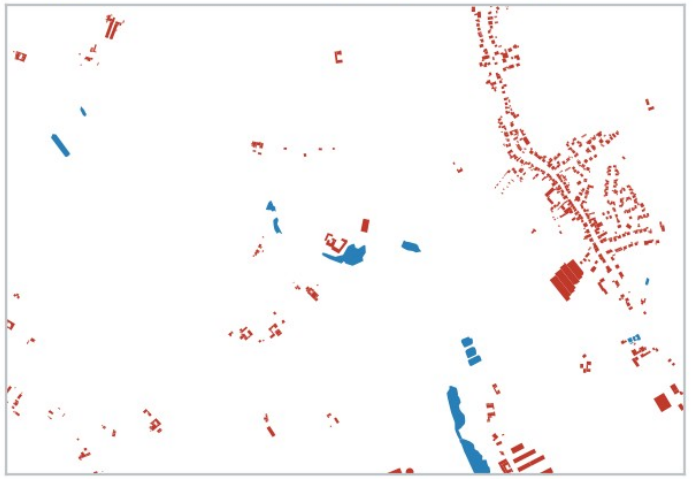
Points: trees (n = 37), zoomed



Lines: roads, streams (n = 190)



Polygons: buildings, ponds (n = 902)



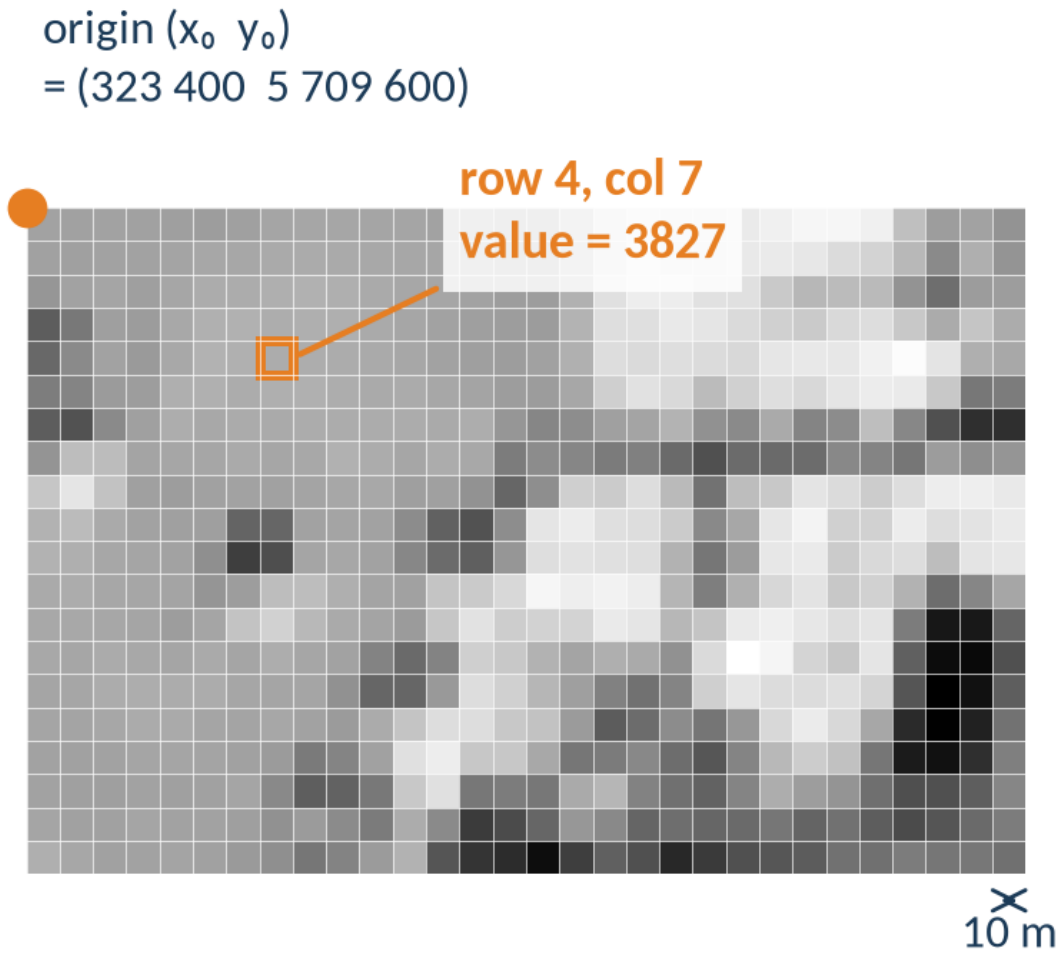
fid	building	name	geometry
40	yes	Freie evangelische G…	POLYGON (…)
1	yes		POLYGON (…)

```
import geopandas as gpd
b = gpd.read_file("kamp_lintfort.gpkg",
                  layer="buildings")
b.geom_type.value_counts()  # Polygon
b.crs                       # EPSG:25832
```

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Raster model: a grid of cells



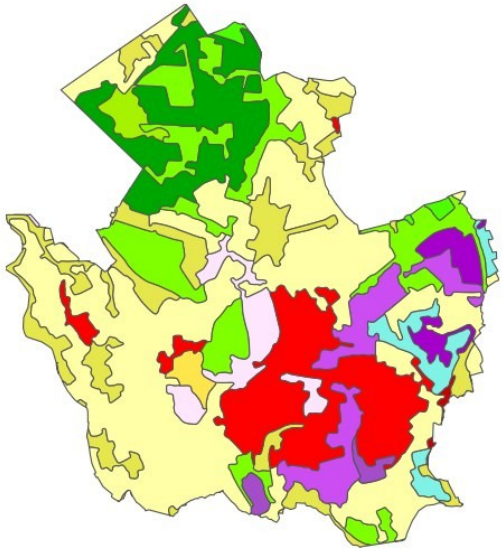
- **Grid:** rows × columns of cells (pixels)
- **Cell size:** resolution, here 10 m × 10 m
- **Bands:** several values per cell (B02 … B08)
- **NoData:** a value that means ‘no measurement’
- **CRS + origin:** place the grid on the Earth

```
import rasterio
src = rasterio.open("s2_l2a_10m.tif")
src.res         # (10.0, 10.0)
src.count       # 4 bands
src.transform   # origin + cell size
src.xy(4, 7)    # row, col -> x, y
src.index(x, y)  # x, y -> row, col
```

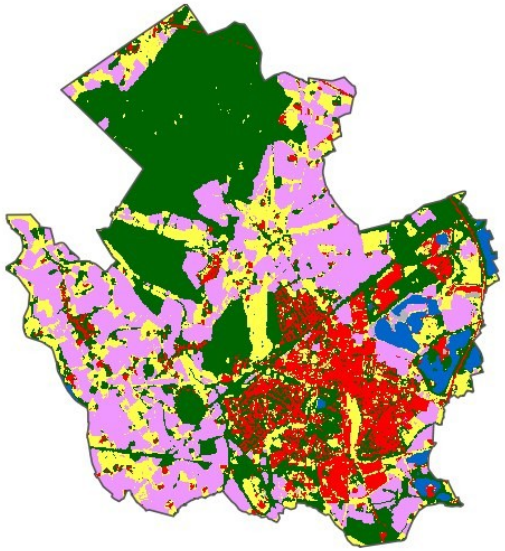
Contains modified Copernicus Sentinel-2 data 2025

Resolution decides what you can see

CORINE 2018: 72 polygons, MMU 25 ha



ESA WorldCover 2021: 631 512 cells of 10 m



231 Pastures
211 Arable land
311 Broad-leaved forest
112 Discontinuous urban
142 Sport/leisure
121 Industrial/commerci
512 Water bodies
131 Mineral extraction
Tree cover
Shrubland
Grassland
Cropland
Built-up
Bare/sparse
Water
Wetland

CORINE Land Cover 2018, © European Union, Copernicus Land Monitoring Service, EEA · ESA WorldCover 10 m 2021 v200 (CC BY 4.0)

Vector or raster?

	Vector	Raster
Models	discrete objects with identity	continuous fields, images
Geometry	points, lines, polygons (exact coordinates)	regular grid, cell size = resolution
Attributes	many columns per feature	one value per cell and band
Strong at	topology, networks, precise boundaries, cartography	map algebra, surfaces, remote sensing
Weak at	continuous variation	sharp boundaries, storage of sparse data
Formats	GeoPackage, GeoJSON, Shapefile, PostGIS, FlatGeobuf	GeoTIFF / COG, NetCDF, Zarr, JPEG2000
Python	shapely, GeoPandas	rasterio, xarray / rioxarray



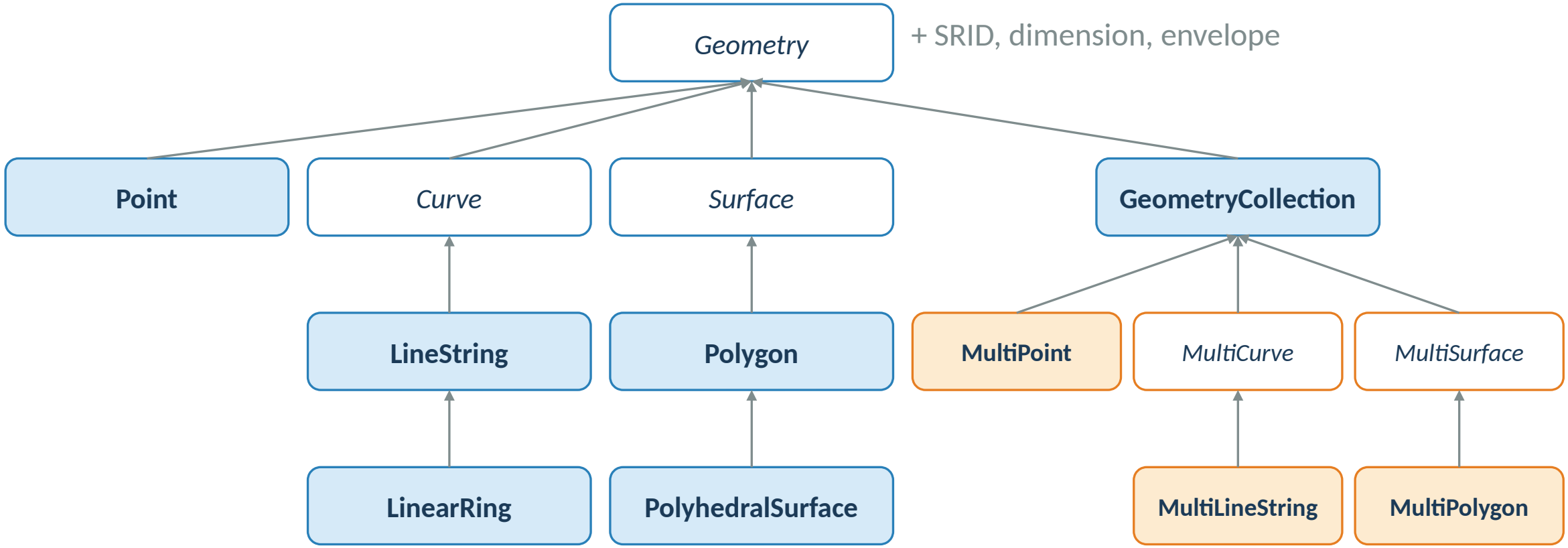
File formats you will meet

Format	Model	Files	Good to know
GeoPackage (.gpkg)	vector + raster	1 (SQLite)	many layers, long field names, OGC standard: our default
Shapefile (.shp)	vector	3–7 (.shp .shx .dbf .prj …)	one layer, field names ≤ 10 characters, 2 GB limit
GeoJSON (.geojson)	vector	1 (text)	web, always WGS 84 lon/lat (RFC 7946), large files get slow
CSV with x/y	points only	1 (text)	no CRS stored: you must know it; keys as text (leading zeros)
GeoTIFF / COG (.tif)	raster	1	bands + georeference; COG = cloud-optimised, read in pieces
NetCDF / Zarr	raster cubes	1 / folder	time series of grids (climate, RADOLAN)

Rule of thumb: GeoPackage for vector exchange, GeoTIFF/COG for rasters, PostGIS when many people work on the same data (geo0840).



OGC Simple Features: the geometry model



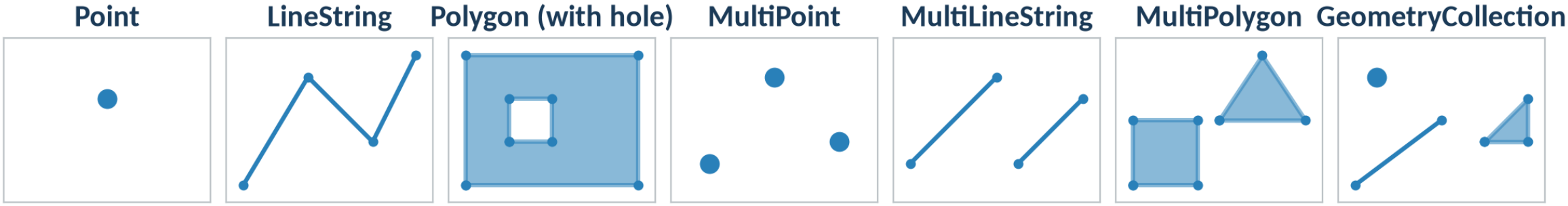
instantiable class

abstract class

Arrows point to the parent class (‘is a’). A Polygon is bounded by LinearRings: one exterior ring, 0…n holes.

Simplified after OGC 06-103r4 Simple Feature Access – Part 1: Common Architecture (ISO 19125-1); redrawn

Seven geometry types, written as WKT

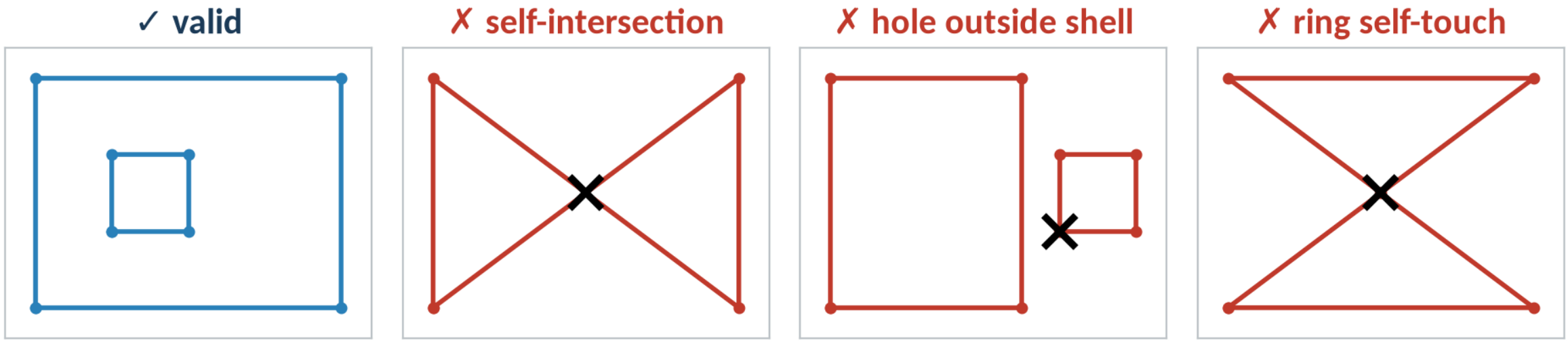



```
POINT (2 2)
LINESTRING (0 0, 1.5 2.5, 3 1, 4 3)
POLYGON ((0 0, 4 0, 4 3, 0 3, 0 0), (1 1, 2 1, 2 2, 1 2, 1 1))
MULTIPOINT ((0.5 0.5), (2 2.5), (3.5 1))
MULTILINESTRING ((0 0.5, 2 2.5), (2.5 0.5, 4 2))
MULTIPOLYGON (((0 0, 1.5 0, 1.5 1.5, 0 1.5, 0 0)), ((2 1.5, 4 1.5, 3 3, 2 1.5)))
GEOMETRYCOLLECTION (POINT (0.5 2.5), LINESTRING (0 0, 2 1.5), POLYGON ((3 1, 4 1, 4 2, 3 1)))
```

```
g = shapely.wkt.loads("POINT (2 2)");  g.wkt, g.wkb_hex
```




Valid geometries: rules of the model

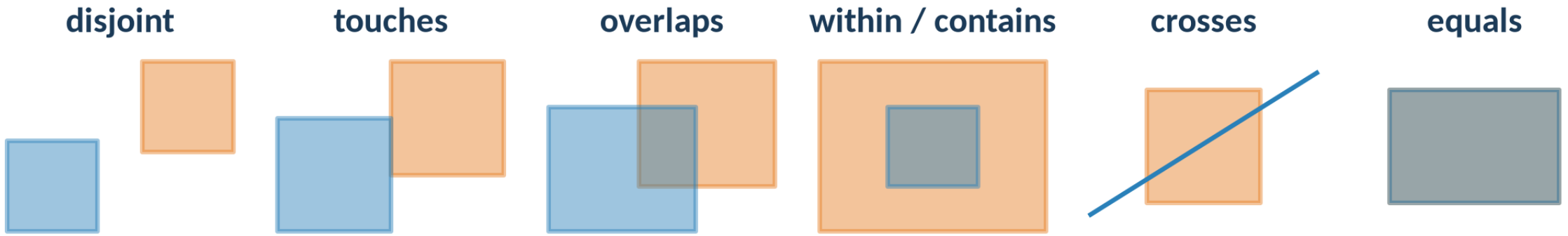


- Rings are closed and do not cross themselves
- Holes lie inside the shell, touching it in one point at most
- Parts of a MultiPolygon do not overlap

```
from shapely.validation import (
    explain_validity, make_valid)
g.is_valid           # False
explain_validity(g)  # Self-intersection
make_valid(g)        # 2 triangles
```




Spatial relationships: DE-9IM



A ∩ B	Interior B	Boundary B	Exterior B
Interior A	2	1	2
Boundary A	1	0	1
Exterior A	2	1	2

under each pair: A.relate(B); ‘overlaps’ = pattern T*T***T**

```
A.within(B), A.touches(B)
A.relate(B)          # '212101212'
A.relate_pattern(B, "T*T***T**")
# whole layers:
gpd.sjoin(wells, gw_bodies,
          predicate="within")
```




Thinking in layers



trees



buildings



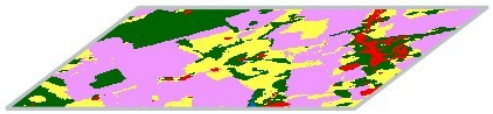
roads



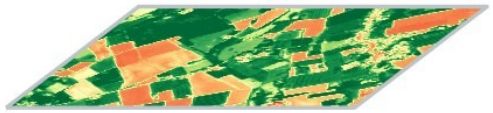
water



landuse



land cover



NDVI



B08 NIR



B04 red

Vector stack: overlay geometries

Raster stack: combine aligned cells

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Take-aways

1. **Two models**
   Vector for objects, raster for fields; the question and the scale decide.

2. **Simple Features**
   One geometry model behind shapely, GeoPandas, PostGIS, GDAL and GeoPackage.

3. **Validity + DE-9IM**
   Check geometries before analysis; spatial predicates are matrix patterns.

4. **Layers**
   Same CRS, same extent: the precondition for overlay and map algebra.



Sources and further reading

- **Standards:** OGC 06-103r4 Simple Feature Access Part 1 (ISO 19125-1); OGC GeoPackage 1.4; DE-9IM: Clementini, Di Felice & van Oosterom (1993)
- **Data:** © OpenStreetMap contributors (ODbL); Contains modified Copernicus Sentinel-2 data 2025, scene S2A_32ULC_20250513 via Earth Search STAC; CORINE Land Cover 2018, © European Union, Copernicus Land Monitoring Service, EEA; ESA WorldCover 10 m 2021 v200 (CC BY 4.0)
- **Further reading (links):** Geo-Python and Automating GIS Processes (Univ. of Helsinki); pythongis.org (Tenkanen, Heikinheimo, Whipp); Shapely and GeoPandas user guides; QGIS Training Manual – vector and raster chapters
- **Figures:** own work, generated with shapely, GeoPandas, rasterio and matplotlib (make_slides.py)

